

Emergency Department KIOSK

Human-Robot Interaction (HRI) Self-Check-in & Escalation

1. PROBLEM SPACE

PROBLEM STATEMENT Patients arriving at the Emergency Department (ED) currently wait in an undifferentiated queue regardless of how their condition is evolving after initial triage. There is no automated way to flag a patient whose self-reported symptoms suggest deterioration before a nurse next does rounds. The kiosk captures a brief self-reported update at check-in and routes a descriptive risk flag to nursing staff for any response above a defined threshold, without replacing nurse rounding or clinical judgement.	
USERS & STAKEHOLDERS PRIMARY: ED PATIENTS Characteristics: Triageed, awaiting rounds; in pain/distressed or limited literacy. Needs: Quick, low-effort symptom change reporting without finding a nurse. SECONDARY: ED NURSES / PORTERS Characteristics: Mixed technical familiarity; nurses need actionable flags. Needs: Reliable escalation path without adding noise to busy unit.	GOALS PRIMARY USERS Short-term: Report symptom change in less than 1 min without waiting. Long-term: Trust genuine deterioration reaches nurse quickly. SECONDARY USERS Short-term: Fewer missed early-deterioration signs between rounds. Long-term: Reduce adverse events from delayed escalation.

2. ETHICAL CONSIDERATIONS INFORMED CONSENT Ensuring patients understand clinical data capture and flag routing before use. PHYSICAL AUTONOMY Accessibility of fixed physical kiosk hardware for immobile or unwell patients. CULTURAL CONTEXT Validating Caribbean ED patient comfort with self-reporting to a machine vs. person.	3. GUIDELINES INTERACTION TYPE Patient-initiated self-report only; no proactive prompting/surveillance. PATIENT/CLINICIAN INTERACTION Patient inputs; nurse receives flag. Kiosk gives zero clinical assessments to patient. ETHICAL DESIGN Consent on first use; severe entries bypass queue to notify nurse immediately.
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4. MINIMUM VIABLE PRODUCT CORE FUNCTION Capture brief self-reported symptom update via touchscreen and route descriptive risk flag to nursing station for entries above severity threshold within 5s. OUT OF SCOPE Vitals-station device integration, voice input, patient-facing risk display. SUCCESS METRIC Flagged entry to nurse acknowledgement less than 5 mins during 2-week pilot; zero adverse events from missed flags.	5 & 6. ENVIRONMENT & FORM PHYSICAL & OPERATIONAL ENVIRONMENT - Fixed kiosk in ED department; shared touchscreen; ward lighting; power instability; ambient noise. - Sequential patient users (no persistence); local flag logging; hospital network only. FORM FACTOR & INTERACTION - Dimensions: Standing kiosk, accessible screen height, matching 3D footprint. - Mobility & Sensors: Stationary furniture (no mobility); touchscreen sensor only. - Communication: Visual/touch only (no audio/voice due to noise & hygiene).
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